

Credit/Lending

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Credit default prediction cho khoản vay nhỏ	Loan application	income, loan amount, employment, repayment history, existing debt	ML classification	probability of default	ML/DS rõ nhất, mentor dễ góp ý	Chủ đề phổ biến, cần explainability để nổi bật
2	Credit risk banding	Sau khi có score	default probability, income stability, debt ratio, repayment history	DS + rule/ML	low / medium / high risk	Dễ hiểu với business/reviewer	Có thể quá đơn giản nếu chỉ threshold
3	Credit explainability / reason codes	Reviewer xem hồ sơ vay	feature importance, SHAP, risk factors	Explainable ML	top risk reasons	Làm credit scoring bớt giống bài Kaggle	Cần hiểu explainability đúng mức
4	Loan approval recommendation	Underwriting decision	risk score, requested amount, income, policy constraints	ML + rule policy	approve / review / reject suggestion	Có workflow ra quyết định rõ	Nhạy cảm, phải nói là decision support

5	Manual review queue prioritization	Reviewer xử lý hồ sơ	risk level, uncertainty, loan amount, missing info	Ranking/ML	review priority	Học human-in-the-loop	Cần giả lập reviewer workload
6	Loan application missing-info detector	Hồ sơ vay chưa đủ dữ liệu	missing fields, inconsistent values, document status	Rule + ML-lite	required additional info	Function nhỏ, thực tế	AI/ML nhẹ nếu đứng riêng
7	Income stability scoring	Đánh giá khả năng trả nợ	salary transactions, income frequency, variance, employer type	DS/time-series	income stability score	Hợp open banking + credit	Cần transaction history synthetic
8	Debt-to-income risk analysis	Underwriting	monthly income, outstanding debt, repayment amount	DS/rule + ML-lite	DTI risk flag	Rất cơ bản nhưng quan trọng	Quá đơn giản nếu không kết hợp model
9	Loan affordability assessment	Trước khi approve	income, expenses, loan installment, household spending	DS + rule/ML	affordable / risky / review	Gần responsible lending	Cần giả lập chi tiêu hợp lý

10	Alternative credit scoring từ transaction data	Thin-file customer lending	wallet/bank transaction history, bill payment, mobile top-up, salary	ML classification	alternative credit score	Rất hợp fintech/financial inclusion	Dữ liệu thật khó, synthetic cần cẩn thận
11	Thin-file borrower scoring	Khách hàng ít lịch sử tín dụng	limited transaction data, KYC, device, payment behavior	ML + uncertainty	risk score + confidence	Bài sâu, thực tế ở fintech	Khó validate nếu thiếu labels
12	Credit score confidence estimation	Model output review	model probability, missing data, feature uncertainty	DS/ML uncertainty	score confidence	Tốt cho human review	Khó hơn scoring thường
13	Loan amount recommendation	Sau khi đánh giá rủi ro	income, repayment capacity, risk band, policy limit	Regression/rule + ML	recommended loan amount	Có output business cụ thể	Nhạy cảm, cần frame là simulation
14	Interest rate / pricing suggestion	Risk-based pricing	risk score, loan term, amount, default probability	ML/rule pricing	suggested rate/tier	Học risk-based pricing	Dễ vượt scope, nhạy cảm
15	Loan term recommendation	Product configuration	income cycle, affordability, risk band, amount	ML/rule recommendation	suggested tenure	Hợp personalization lending	Cần define objective rõ

16	Loan purpose classification	Loan application text	loan purpose description, category, amount	NLP classification	purpose category	Nhỏ, dễ làm	Ít giá trị nếu đứng riêng
17	Fraudulent loan application anomaly	Loan application screening	repeated device, identity overlap, suspicious income, duplicate docs	Anomaly/rule/ML	suspicious application flag	Gần risk, cụ thể	Lẫn với fraud/KYC, cần scope rõ
18	Loan repayment late-warning	Khoản vay đang active	repayment schedule, days before due, past delays, balance trend	ML classification/time-series	late repayment risk	Hẹp hơn credit scoring, thực tế	Cần synthetic loan lifecycle data
19	Early delinquency prediction	Sau kỳ trả đầu tiên	first repayment behavior, partial payment, due-date behavior	ML classification	early delinquency probability	Rất thực tế trong lending	Need labels/synthetic scenarios
20	Overdue severity prediction	Khi đã overdue	days past due, contact history, partial repayment, amount	ML classification/regression	mild / severe overdue	Hữu ích cho collection	Sensitive, cần ethical framing

21	Repayment behavior segmentation	Portfolio monitoring	payment timeliness, partial payments, prepayments	Clustering/DS	borrower behavior segment	Học DS segmentation	Khó biết segment đúng/sai
22	Prepayment prediction	Loan monitoring	early repayments, income inflow, interest rate, tenure	ML classification	prepayment probability	Có business angle portfolio	Ít phổ biến với intern scope
23	Loan restructuring eligibility support	Khách hàng khó trả nợ	overdue status, income drop, repayment history, hardship signal	Rule + ML-lite	eligible for restructuring / review	Có responsible lending angle	Domain/policy phức tạp
24	Collection prioritization	Debt collection queue	overdue days, amount, repayment likelihood, contact history	ML ranking/segmentation	priority list	Gần lending ops thật	Nhạy cảm, phải tránh aggressive collection framing
25	Best contact time/channel prediction	Collection/customer contact	past contact success, time, channel, customer segment	ML classification/ranking	best time/channel	Function nhỏ, thực tế	Cần contact outcome data synthetic
26	Collection message personalization	Nhắc nợ/repayment reminder	overdue severity, customer segment, prior response	LLM templating + guardrails	message draft	Hợp AI, nhưng cần ethical guardrails	Dễ nhạy cảm, cần tone an toàn

27	Promise-to-pay reliability scoring	Collection follow-up	customer promise, past promises kept, payment delay	ML classification	promise reliability score	Bài rất cụ thể, ops-oriented	Data khó, nhạy cảm
28	Loan portfolio risk dashboard	Portfolio monitoring	default rate, overdue buckets, risk bands, product type	DS analytics	portfolio risk summary	Dễ demo, business-friendly	Ít ML nếu đứng riêng
29	Vintage analysis for loans	Portfolio analytics	loan cohort by origination month, default over time	DS/cohort analysis	vintage curves	Học credit analytics thật	Có thể quá DS/reporting
30	Cohort-based default trend detection	Portfolio monitoring	cohort default rate, product/channel/time	DS/anomaly	unusual default trend alert	Có monitoring angle	Need synthetic time data
31	Credit model drift monitoring	Model governance	feature distribution, prediction distribution over time	DS drift metrics	drift alert	Tư duy AI governance tốt	Nên làm add-on, không làm chính
32	Credit model performance monitoring	Sau khi có actual repayment labels	prediction, actual default, time	DS metrics	AUC/F1/calibration trend	Rất tốt nếu có model chính	Need delayed labels giả lập

33	Credit decision fairness dashboard	Model governance	approval rate, default rate, error rate by segment	DS fairness metrics	fairness report	Có responsible AI angle	Nhạy cảm; cần dùng synthetic groups
34	Human override analysis	Reviewer khác model	model score, reviewer decision, actual outcome	DS analytics	override pattern report	Rất hay cho decision support	Cần mô phỏng reviewer decisions
35	Credit decision audit trail	Compliance logging	model version, input features, score, reviewer, decision	SE + DS audit	audit record	Hợp fintech compliance	Không phải ML core
36	Adverse action reason generation	Sau reject/review	top risk factors, policy reason, applicant data	LLM structured explanation + rules	reject/review reason draft	Hợp AI + explainability	Cần tránh legal/ethical overclaim
37	Loan officer assistant for case review	Reviewer đọc hồ sơ	customer profile, score, documents, history	LLM summary + evidence	case summary + questions	Hợp AI assistant có context	Nếu không có evidence thì thành chatbot
38	Policy compliance checker for loan decision	Sau decision	policy rules, applicant data, decision	Rule + LLM judge	possible policy violation	Có governance angle	LLM judge dễ sai, cần rule base

39	P2P borrower risk scoring	Borrower listing	borrower profile, loan purpose, requested amount, history	ML risk score	borrower risk band	Bám sandbox P2P lending	Data khó, legal/domain nhạy cảm
40	P2P lender-borrower matching	Marketplace recommendation	borrower risk, lender preference, amount, term	Ranking/recommendation	suitable loan listings	Kết hợp lending + recommendation	Dễ mơ hồ nếu thiếu objective
41	P2P loan listing quality scoring	Listing review	purpose text, completeness, borrower info, risk score	NLP + rule/ML	listing quality score	Function cụ thể cho P2P	Cần synthetic listing text
42	Borrower explanation assistant for P2P lenders	Lender xem listing	risk score, borrower profile, loan purpose	LLM structured explanation	readable risk explanation	Hợp AI + lending	Cần tránh tạo cảm giác financial advice thật
43	BNPL repayment risk scoring	Buy-now-pay-later checkout	cart value, user history, repayment behavior, merchant	ML classification	BNPL approval/risk score	Rất fintech thực tế	Data synthetic là chính

44	Credit limit increase recommendation	Existing borrower/customer	repayment history, usage, income stability, default risk	ML/rule recommendation	increase / keep / decrease limit	Hợp digital lending/cards	Nhạy cảm, cần frame prototype
45	Loan product recommendation	Cross-sell/personalization	persona, income, transaction pattern, need signal	Recommendation/ranking	suggested loan product	Bám personalization	Dễ marketing, cần risk guardrail
46	Financial hardship signal detection	Existing borrower monitoring	income drop, spending drop, missed payments, support tickets	DS/anomaly/ML	hardship risk flag	Có responsible/ethical angle	Cần cẩn thận về privacy/ethics
47	Early warning from transaction behavior	Before delinquency	salary missing, cashflow drop, unusual spending, bill missed	DS/anomaly + ML	repayment stress alert	Kết hợp open banking + lending	Cần transaction data synthetic
48	Loan schedule anomaly detector	Loan servicing	repayment schedule, interest, principal, due dates	Rule + anomaly	schedule inconsistency	Học loan lifecycle	Nặng domain, AI nhẹ

49	Repayment allocation validator	Payment to loan	amount split to principal, interest, fee, penalty	Rule + anomaly	allocation error alert	Học core banking sâu	Accounting/loan domain khó
50	Loan document completeness checker	Loan application docs	required docs, extracted fields, missing signatures	LLM/OCR + validation	missing doc checklist	Hợp AI/document workflow	Có thể trùng KYC/document

Package 1 — Credit Risk Decision Support Prototype

Gồm:

- #1 Credit default prediction
- #2 Credit risk banding
- #3 Credit explainability / reason codes
- #4 Loan approval recommendation
- #35 Credit decision audit trail
- #36 Adverse action reason generation

Package 2 — Loan Early Warning & Repayment Risk

Gồm:

- #18 Loan repayment late-warning
- #19 Early delinquency prediction
- #20 Overdue severity prediction

#21 Repayment behavior segmentation
#47 Early warning from transaction behavior

Package 3 — Alternative Credit Scoring from Transaction Data

Gồm:

#7 Income stability scoring
#9 Loan affordability assessment
#10 Alternative credit scoring
#11 Thin-file borrower scoring
#47 Early warning from transaction behavior

Package 4 — Collection Intelligence Prototype

Gồm:

#24 Collection prioritization
#25 Best contact time/channel prediction
#26 Collection message personalization
#27 Promise-to-pay reliability scoring
#46 Financial hardship signal detection

Package 5 — Credit AI Governance Add-on

Gồm:

#31 Credit model drift monitoring

- #32 Credit model performance monitoring
- #33 Credit decision fairness dashboard
- #34 Human override analysis
- #35 Credit decision audit trail